# Ganesh Gupta

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## Experience

**Oracle** | *Software Engineer, AI Platform* Jun 2025 – Present

- Provisioned and operated **GPU clusters** on **AWS** for distributed training and inference, tuning **transport-layer performance** and placement so jobs held throughput under heavy churn and hardware scarcity.
- Wrote systems-level **scheduling, resource-allocation, and fault-tolerant placement** code that reschedules work around failed nodes, keeping training and inference workloads running at **99%+ uptime**.
- Built a pluggable **distributed executor** routing workloads to Docker and **AWS Lambda**, backed by a content-addressed **S3** artifact store with memoization at **sub-200ms p99**, eliminating redundant recomputation.
- Built observability and autoscaling for **vLLM inference serving** with **Prometheus/Grafana**, and trained and fine-tuned **PyTorch** models on **multi-GPU clusters**.

**QureAI** | *ML Infrastructure Engineer* Mar 2026 – Present

- Built a **durable execution pipeline** with **remote execution** routed across **Docker** and **AWS** environments and per-step **Redis checkpointing**, so interrupted workflows resume from the last committed step instead of rerunning.
- Cut **p99 latency from 420ms to 140ms** across **50K+ daily requests** by migrating to **async serving** with horizontal **autoscaling** and load-aware routing.
- Built production **observability** tracking latency, drift, and regressions across deployments, with failure alerting and automated **rollback** that cut manual incident-response toil.

**Nomura Research Institute Financial Technologies** | *Software Engineer* Sep 2019 – Aug 2023

- Built and operated **25+ production Python and Java** distributed services for **1M+ users** at **99.9% uptime** on AWS (EC2, RDS, SQS).
- Cut production escalations **60%** in one quarter by standardizing **retry, timeout, and failure-propagation** semantics across **25+ services**, and drove **root-cause analysis** and post-incident reviews.
- Cut nightly aggregation blocking downstream settlement from **8 minutes to 90 seconds** by redesigning **SQL and PL/SQL** with indexed views and batch partitioning, removing lock contention.
- Led **architecture and code reviews** across the full development lifecycle, mentoring engineers on **design patterns** and scaling.

## Projects

**VPC Control Plane Simulator / Distributed KV Store** `github.com/ganeshhgupta/distributed-kv-store`

- **VPC control plane simulator** modeling **subnet routing**, traffic forwarding, and **security group enforcement**, alongside a **Raft**-consensus key-value store with linearizable reads and log compaction.

**RootCause AI** `github.com/ganeshhgupta/rootcause-ai`

- Multi-agent system that automates **production incident diagnosis**, tracing system-level failures across distributed services to a structured **root-cause** report.

**Watch-your-LLM** `github.com/ganeshhgupta/watch-your-llm`

- Observability platform tracking **latency** and regressions across model versions, with FastAPI, PostgreSQL, and **CI/CD**.

## Education

**University of Texas at Arlington** | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

**MCKV Institute of Engineering** | *B.Tech Computer Science* Aug 2016 – Jul 2020

## Skills

**Languages:** Python, SQL, Java

**ML Systems:** PyTorch, vLLM, TensorRT-LLM, CUDA, Distributed Training and Inference, Profiling, Benchmarking, Performance Optimization

**AI Infrastructure:** GPU Clusters, InfiniBand, Slurm, Fault-Tolerant Scheduling, Autoscaling, Prometheus/Grafana, Observability

**Platforms:** AWS (EC2, S3, Lambda, SQS, SageMaker), GCP Vertex AI, Docker, Kubernetes, CI/CD

**Certifications:** AWS Certified Developer Associate, GCP Professional MLOps Engineer